A two-spectrum sextic criterion for the complex Weinberg 3HDM

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This manuscript integrates the analytic argument with the confirmed errata and the F002 finite-input formal construction. Verification scope is stated in Section 9; a declaration-level concordance accompanies this manuscript. “Exact” below refers to algebraic equations, root selection, signs and limits; the accompanying floating-point prototype is not an exact decision program.

1. Statement, normalization and geometric core

For three columns $\Phi_i \in \mathbb{C}^2$, put $K_{ij} = \Phi_i^\dagger \Phi_j$, $r_i = K_{ii}$, and consider

$$f(K) = \frac{1}{2}\sum_{i=1}^{3} a_i r_i^2 + \sum_{i<j}\{b_{ij} r_i r_j + c_{ij}|K_{ij}|^2 + \mathrm{Re}(d_{ij}K_{ij}^2)\}. \tag{1}$$

The nine coefficients a_i, b_{ij}, c_{ij} are real and the three d_{ij} are arbitrary complex numbers. Pair order is always $12, 13, 23$. No restriction to real couplings, nonzero couplings or neutral fields is made.

Homogeneity reduces the question to the normalized Gram domain

$$\mathcal{D} = \{K = K^\dagger \succeq 0 : \mathrm{tr}\, K = 1, \mathrm{rank}\, K \le 2\}, \qquad m = \min_{K\in\mathcal{D}} f(K). \tag{2}$$

This is exactly the physical domain: every such matrix has a two-row Gram factor. Rank one represents aligned, electrically neutral doublets; rank two represents charge-breaking directions. Missing doublets occur as support faces.

The geometric core is the one-dimensional kernel of a rank-two Gram matrix. A charged stationary point can be reconstructed from a rank-one projector onto that kernel. Its kernel problem has the same phase-current algebra as a neutral configuration, with transformed couplings and a weighted normalization:

The two branches are: rank-one Gram matrices give the primal neutral spectrum; rank-two Gram matrices give a kernel projector and the inverse-operator spectrum. Both use the same sextic phase engine. Physical filtering, finite minimization, and successive bounded residues then recover the minimum and a field witness.

The Gram-domain and constrained-stationarity framework is established prior work, not a novelty claim here [MN15]. The new proposed reduction is the explicit common sextic engine, its inverse-operator charged reconstruction, and their complete algebraic treatment of exceptional inputs.

Theorem (universal exact criterion)

Apply the fifteen-coordinate positive infinitesimal deformation in Section 7 to (1). Apply the explicit spectrum construction of Sections 2–4 to the deformed couplings, in the indicated real closed ordered field. Let $\mathcal{C}_\epsilon$ be the retained candidate values and $m_\epsilon = \min \mathcal{C}_\epsilon$. There are at most thirty distinct retained values, before removal of duplicate values and after identifying the discrete doublet-sign copies. The set is nonempty. Then

$$\begin{aligned} f \ge 0 \text{ on } \mathcal{D} &\iff m_\epsilon > 0, \\ f > 0 \text{ on } \mathcal{D} &\iff m_\epsilon > 2\epsilon_0, \\ m < 0 &\iff m_\epsilon < 0, \\ m = 0 &\iff 0 < m_\epsilon < 2\epsilon_0, \\ m &= \mathrm{st}(m_\epsilon). \end{aligned} \tag{3}$$

Here signs are exact ordered-field signs, not tests at a small floating-point number. A minimizing retained matrix has a bounded successive residue K_0 with $K_0 \in \mathcal{D}$ and $f(K_0) = m$. Section 8 gives a terminating algebraic residue procedure and an explicit two-row Gram factor. Thus a negative or zero field direction is recovered when applicable.

For ordinary coefficients in a nonempty Zariski-open set, the same two spectra can be used without deformation: their physical minimum is exactly m. The deformation is the unconditional prescription, including all singular operators, real-coupling loci, flat directions, vanishing couplings, stationary continua and denominator degeneracies. It is not a numerical regularization.

The finite construction is relative to exact arithmetic, equality/order, and finite real-root interfaces over the ordered tower. F002 proves uniform input-derived syntax and its semantic refinement (Section 8.1); it supplies no verified rational/algebraic backend or extracted runnable solver. Rational or algebraic input permits standard effective representations in principle, but their implementation and verification are additional work. No computability claim is made for arbitrary uncomputable real numbers.

2. The common weighted spectrum

Input data are $\alpha_i, e_{ij} \in F$, $d_{ij} \in F(i)$, and $g_i \in F$, where F is real closed. Temporarily assume generic data in the precise sense justified in Section 6. In particular all g_i are nonzero.

For ordinary real phases set $q_i = e^{i\theta_i}$, $q_1 = 1$. Over a general real closed field, q_i means a point on the algebraic unit circle, not a transcendental exponential. Set

$$B_{ii} = \alpha_i/2, \qquad B_{ij} = \frac{1}{2}[e_{ij} + \mathrm{Re}(d_{ij}q_j\bar{q}_i)], \quad i < j,$$

$$Q(r,q) = \frac{r^T B(q) r}{(g^T r)^2}, \qquad r_i \geq 0, \quad \mathbf{1}^T r = 1, \quad g^T r \neq 0. \tag{4}$$

The procedure below produces every projective stationary point of (4), with support recorded, up to sign copies of the underlying complex amplitudes. Retain all branches, not only the smallest value of this auxiliary quotient. Negative $g^T r$ is allowed.

2.1 One-element supports

For each i, include

$$r = e_i, \qquad \nu = \frac{\alpha_i}{2g_i^2}. \tag{5}$$

Inactive phases are irrelevant.

2.2 Two-element supports

For every pair $i < j$ and each $\sigma \in \{-1, +1\}$, set

$$q_j\bar{q}_i = \sigma\frac{\bar{d}_{ij}}{|d_{ij}|}, \quad A_i = \alpha_i/2, \quad A_j = \alpha_j/2, \quad h = \frac{e_{ij} + \sigma|d_{ij}|}{2},$$

$$n = (A_j g_i - h g_j,\ A_i g_j - h g_i)^T, \quad W = A_j g_i^2 - 2h g_i g_j + A_i g_j^2. \tag{6}$$

If the displayed denominators are nonzero, set

$$(r_i, r_j) = n/(n_i + n_j), \quad r_k = 0, \qquad \nu = \frac{A_i A_j - h^2}{W}. \tag{7}$$

Retain only strictly positive r_i, r_j and $g^T r \neq 0$. There are at most six such candidates. Do not divide by $A_i A_j - h^2$: zero values are legitimate. Generic data ensure $d_{ij} \neq 0$ and $W \neq 0$; Section 7 covers their failures.

To prove coverage, phase stationarity on a two-element support gives (6). Radial stationarity of the homogeneous quotient gives $B_J r_J = \nu(g_J^T r_J) g_J$. The adjugate of the two-by-two matrix yields (7). This argument does not assume that B_J is invertible.

2.3 Full support: currents eliminate amplitudes

Define

$$s_{12} = \operatorname{Im}(d_{12} q_2), \quad s_{13} = \operatorname{Im}(d_{13} q_3), \quad s_{23} = \operatorname{Im}(d_{23} q_3 \bar{q}_2), \quad j = (s_{23}, -s_{13}, s_{12})^T. \tag{8}$$

The two phase derivatives vanish exactly when

$$r_1 r_2 s_{12} = r_2 r_3 s_{23} = -r_1 r_3 s_{13}. \tag{9}$$

If all $r_i > 0$ and

$$\operatorname{Im}(d_{12} d_{23} \bar{d}_{13}) \neq 0, \tag{10}$$

the common current in (9) cannot vanish. Indeed vanishing forces every $d_{ij} q_j \bar{q}_i$ to be real, whereas their invariant loop product is $d_{12} d_{23} \bar{d}_{13}$. Consequently

$$r = \frac{j}{\mathbf{1}^T j}. \tag{11}$$

This eliminates the radial variables before polynomial elimination.

Let

$$C = \begin{pmatrix} e_{13} d_{12} & -e_{12} d_{13} & \alpha_1 d_{23} - d_{13} \bar{d}_{12} \\ e_{23} d_{12} & d_{12} d_{23} - \alpha_2 d_{13} & e_{12} d_{23} \\ \alpha_3 d_{12} - d_{13} \bar{d}_{23} & -e_{23} d_{13} & e_{13} d_{23} \end{pmatrix}, \qquad z = (q_2, q_3, q_3 \bar{q}_2)^T. \tag{12}$$

A direct identity is

$$2B(q) j(q) = \operatorname{Im}(Cz). \tag{13}$$

For example, the cross term in the first component follows from $\operatorname{Re} v \operatorname{Im} u - \operatorname{Re} u \operatorname{Im} v = \operatorname{Im}(u\bar{v})$. The other two components follow by the same complex multiplication. The full identity has also been checked as a polynomial identity with every coupling symbolic, independently of numerical stationary-point searches.

Put

$$h^{(2)} = g_1 C_{2\bullet} - g_2 C_{1\bullet}, \qquad h^{(3)} = g_1 C_{3\bullet} - g_3 C_{1\bullet}. \tag{14}$$

The remaining radial conditions are precisely

$$\operatorname{Im}(h^{(2)} z) = 0, \qquad \operatorname{Im}(h^{(3)} z) = 0. \tag{15}$$

Indeed stationarity of (4) requires $Br = \nu(g^T r) g$; (11) and (13) reduce this to proportionality of $\operatorname{Im}(Cz)$ and g. Conversely, (15), positive (11), and $g^T r \neq 0$ imply this proportionality. Multiplication by r^T determines its coefficient and gives exactly (4). Thus the retained roots are actual stationary points, not merely solutions of necessary equations.

2.4 Two biquadratics and a sextic

Use projective half-angle coordinates

$$q_2 = \frac{1 - t^2 + 2it}{1 + t^2}, \qquad q_3 = \frac{1 - u^2 + 2iu}{1 + u^2}. \tag{16}$$

For any complex row (A, B, C), define the explicit real polynomial

$$\begin{aligned} P_{A,B,C}(t,u) = {} & [\operatorname{Im} A(1 - t^2) + 2 \operatorname{Re} A\, t](1 + u^2) \\ & + [\operatorname{Im} B(1 - u^2) + 2 \operatorname{Re} B\, u](1 + t^2) \\ & + \operatorname{Im} C[(1 - t^2)(1 - u^2) + 4tu] \\ & + 2 \operatorname{Re} C[u(1 - t^2) - t(1 - u^2)]. \end{aligned} \tag{17}$$

Then (15) is exactly $F(t,u) = G(t,u) = 0$, where $F = P_{h^{(2)}}$, $G = P_{h^{(3)}}$. Each has bidegree at most $(2,2)$. Both have the fixed nonreal base points (i,i) and $(-i,-i)$. Therefore

$$S(t) = \frac{\mathrm{Res}_u(F,G)}{1+t^2} \tag{18}$$

is an exact polynomial quotient of degree at most six. The numerator is the ordinary four-by-four Sylvester determinant of two quadratics in u. No physical real phase is removed by the denominator in (18).

Bihomogenize both polynomials to bidegree $(2,2)$ in $([T_0 : T_1], [U_0 : U_1])$. Use all four affine charts, obtained by reversing the coefficient order in t, u, or both. In every chart isolate all real roots of (18), specialize the two quadratics at each root, and isolate all real roots of their gcd. Keep only common roots of the original two equations. This handles vanishing leading coefficients and phases equal to π. The universal base factor remains $1+t^2$ in each chart; in a reciprocal chart the associated nonreal u-coordinate can change sign.

If the two bihomogeneous forms have no common component, they have at most eight complex intersections counted with multiplicity; the two fixed nonreal points leave at most six physical full-support points. Equivalently, their homogeneous resultant is a binary octic with the fixed quadratic factor. Multiplicities of projected common roots account for multiple roots in a fiber. This is only a degree/coverage bound, not an assumption that the six roots are real or simple.

For each real root pair compute (8) and (11). Discard a zero denominator, a nonpositive component, or $g^T r = 0$. For each remaining pair include $\nu = Q(r,q)$. Form a rank-one projector by choosing any algebraic square roots

$$w_i^2 = r_i q_i, \qquad P_{ij} = \bar{w}_i w_j. \tag{19}$$

Changing square-root signs conjugates P by a diagonal sign matrix and changes none of the tests or values below. The common weighted spectrum contains at most $3+6+6 = 15$ values, with supports and projectors retained.

3. The operator and the two input spectra

On Hermitian three-by-three matrices use $\langle X, Y\rangle = \mathrm{tr}(XY)$. Let A have diagonal a_i and off-diagonal entries b_{ij}. Define the real self-adjoint map

$$(LX)_{ii} = (A \,\mathrm{diag}\, X)_i, \qquad (LX)_{ij} = c_{ij}X_{ij} + \bar{d}_{ij}\bar{X}_{ij}. \tag{20}$$

Then $f(K) = \langle K, LK\rangle/2$. Each off-diagonal real block is

$$\begin{pmatrix} c + \mathrm{Re}\, d & -\mathrm{Im}\, d \\ -\mathrm{Im}\, d & c - \mathrm{Re}\, d \end{pmatrix}.$$

For invertible L, set $R = L^{-1}$, $G = A^{-1}$, and $\Delta_{ij} = c_{ij}^2 - |d_{ij}|^2$. Explicitly

$$(RX)_{\mathrm{diag}} = G \,\mathrm{diag}\, X, \qquad (RX)_{ij} = \frac{c_{ij}X_{ij} - \bar{d}_{ij}\bar{X}_{ij}}{\Delta_{ij}}. \tag{21}$$

Set $g = G\mathbf{1}$, $\beta = \mathbf{1}^T G\mathbf{1}$. In particular $RI = \mathrm{diag}(g)$.

Primal input. Run Section 2 with

$$(\alpha, e, d, g) = (a, b+c, d, \mathbf{1}). \tag{22}$$

Every retained projector is a neutral physical matrix and its spectral value is exactly $f(P)$. Include all of these values in $\mathcal{C}$.

Dual input. Run the same procedure with

$$\alpha'_i = 2G_{ii}, \quad e'_{ij} = 2\left(G_{ij} + \frac{c_{ij}}{\Delta_{ij}}\right), \quad d'_{ij} = -\frac{2d_{ij}}{\Delta_{ij}}, \quad g' = G\mathbf{1}. \tag{23}$$

For each resulting projector P, put

$$a_P = \langle P, RI\rangle = g^T r, \quad p_P = \langle P, RP\rangle, \quad D_P = \beta p_P - a_P^2. \tag{24}$$

The dual quotient is exactly $\nu = p_P/a_P^2$, including its complex phases. Discard $D_P = 0$ and construct

$$K_P = \frac{p_P RI - a_P RP}{D_P}, \qquad E_P = \frac{p_P}{2D_P} = \frac{\nu}{2(\beta\nu - 1)}. \tag{25}$$

Do not divide by p_P or ν, and do not require either to be nonzero. The projective-stationarity identities prove $K_P P = 0$ and $\operatorname{tr} K_P = 1$. The remaining physical test is Section 4.

4. A scalar charged-admissibility test

Define

$$\beta_2 = \langle RI, RI\rangle, \quad a_{2,P} = \langle RI, RP\rangle, \quad p_{2,P} = \langle RP, RP\rangle,$$
$$\chi_P = \frac{p_P^2\beta_2 - 2p_P a_P a_{2,P} + a_P^2 p_{2,P}}{D_P^2}. \tag{26}$$

Then $\chi_P = \operatorname{tr} K_P^2$. Since K_P is Hermitian, has trace one and already has a nontrivial kernel, its eigenvalues are $(0, x, 1-x)$. Hence

$$K_P \succeq 0 \quad\Longleftrightarrow\quad \chi_P \le 1. \tag{27}$$

Retain exactly those dual candidates passing (27), adding their values E_P to $\mathcal{C}$. Equality may produce a rank-one duplicate and is retained. Purity alone would not enforce the physical rank bound: $I/3$ is a counterexample unless the separate kernel identity is proved.

All scalars in (26) can be evaluated without choosing half-angle square roots. Namely $\beta_2 = \|G\mathbf{1}\|^2$, $a_{2,P} = (G^2\mathbf{1})^T r$, and $p_{2,P}$ is the neutral numerator with coefficients

$$\alpha_i'' = 2(G^2)_{ii}, \quad e_{ij}'' = 2\left((G^2)_{ij} + \frac{c_{ij}^2 + |d_{ij}|^2}{\Delta_{ij}^2}\right), \quad d_{ij}'' = -\frac{4c_{ij}d_{ij}}{\Delta_{ij}^2}. \tag{28}$$

This follows by squaring each real two-by-two inverse block. Matrix reconstruction and principal-minor tests provide redundant implementation checks.

5. Charged-to-neutral equivalence and global coverage

5.1 Stationarity in two representations

A rank-two point of $\mathcal{D}$ lies on a smooth rank-two stratum. The normal space to fixed trace and rank is spanned by I and its kernel projector P. Thus stationarity implies

$$LK = \lambda I + \mu P. \tag{29}$$

This can independently be derived in the original doublet variables. Write $K = \Phi^\dagger\Phi$ with Φ a rank-two two-by-three matrix. The real field derivative of f is $2\Phi LK$, while that of $\operatorname{tr} K$ is 2Φ. Lagrange stationarity gives $\Phi LK = \lambda\Phi$. Hermiticity then forces $LK - \lambda I$ to be supported in the one-dimensional kernel of Φ, which gives (29). Taking the trace against K fixes $\lambda = 2f(K)$. The field-gradient identity has been checked for all twelve independent real field coordinates with all fifteen couplings symbolic.

Applying R, taking the trace, and taking the expectation in P yields

$$\beta\lambda + a_P\mu = 1, \qquad a_P\lambda + p_P\mu = 0. \tag{30}$$

For $D_P \neq 0$ this is exactly (25), with $\lambda = p_P/D_P$, $\mu = -a_P/D_P$.

For $a_P \neq 0$, write $P = zz^\dagger$, $z^\dagger z = 1$. Projective stationarity of p_P/a_P^2 is equivalent to

$$(a_P RP - p_P RI)z = 0. \tag{31}$$

For completeness, its directional derivative is $4\,\mathrm{Re}[\delta z^\dagger (a_P RP - p_P RI)z]/a_P^3$. The expectation of the vector in (31) against z vanishes identically, so the normalization multiplier is zero. Equation (31) is exactly $K_P z = 0$ from (25). Thus the spectral stationary equations and the charged kernel condition are equivalent, not only one-way implications.

Missing components of z cause no omitted first derivatives. If $z_i = 0$, then $P_{ij} = z_i \bar{z}_j = 0$ for every j. For $j \neq i$, (21) therefore gives $(RP)_{ij} = 0$, but its diagonal entry $(RP)_{ii} = (Gr)_i$ may be nonzero. The required vector identity is nevertheless

$$((RP)z)_i = (Gr)_i z_i + \sum_{j \neq i} (RP)_{ij} z_j = 0, \qquad ((RI)z)_i = g_i z_i = 0. \tag{31a}$$

Thus the off-support components of (31) vanish. The same calculation using (20) gives $((LP)z)_i = 0$ and $((LI)z)_i = 0$: diagonal mixing is again multiplied by $z_i = 0$. The on-support radial and phase derivatives exhaust the remaining tangent directions; at a missing complex amplitude the real and imaginary first variations vanish by these vector identities. This is the support property used in the proof, not preservation of an entire zero row. It relies on the individual sign-flip structure and need not hold for a general 3HDM. The inverse identity is also the literal conclusion of W3.Operator.inverse_projector_zero_component in F002.

5.2 Exceptional charged equations

The following exceptions must not be silently discarded.

If $a_P = 0$ but $p_P \neq 0$, (30) forces $\mu = 0$ and $K = RI/\beta$, which has rank three when every $g_i \neq 0$. It cannot be our rank-two point. If $D_P = 0$, consistency of (30) forces $a_P = p_P = 0$, $\beta \neq 0$, and $\lambda = 1/\beta$. Moreover $\mu \neq 0$ for the same rank reason, and the kernel equation implies

$$RPz = -\frac{\lambda}{\mu} RIz. \tag{32}$$

It follows that P is a stationary zero of p_P restricted to $a_P = 0$ on projective space. Section 6 proves that these stationary zeros are absent for algebraically generic weighted data. They need not be absent for arbitrary real data; an exact nonreal-phase counterexample is in DE RIVATIONS.md. Section 7, not an unproved nonzero-denominator assumption, covers these cases.

5.3 Minimum coverage

A global minimum of (2) has rank one or two. Rank one is covered by the primal spectrum on its actual support. A rank-two minimum is stationary on its smooth stratum, hence obeys (29). For generic coefficients Sections 5.1–5.2 place it in the dual spectrum and it passes the exact physical filter (27). Conversely, every retained matrix is in $\mathcal{D}$ and has the stated energy. Therefore

$$m = \min \mathcal{C} \tag{33}$$

for generic data. This proves both directions of the global result; numerical absence of other minima is not used.

Over a real closed extension, use the semialgebraic extreme-value theorem for closed bounded domains, equivalently first-order transfer of the ordinary real compact minimum assertion. Topological compactness of a non-Archimedean field is not being assumed. The stationary equations are algebraic circle equations, so no exponential or trigonometric function over that field is needed.

6. Why generic data really avoid the missing branches

Only a proper algebraic exceptional set is removed here. Section 7 subsequently removes this qualification from the theorem.

6.1 Elementary exclusions and the full-support intersection

Invertibility of A, all $\Delta_{ij} \neq 0$, all $g_i \neq 0$, $\beta \neq 0$, nonzero d_{ij} and the nonreal loop condition each exclude proper polynomial zero sets after clearing denominators. Pair-face $W = 0$ also does so: eliminating $|d|$ gives, for a pair,

$$[A_j g_i^2 - e_{ij} g_i g_j + A_i g_j^2]^2 - |d_{ij}|^2 g_i^2 g_j^2 = 0. \tag{34}$$

This polynomial is not identically zero.

For the two full-support bihomogeneous forms, having a common factor of some positive bidegree is a closed algebraic condition in their projective coefficient spaces: there are finitely many possible factor bidegrees, and multiplication maps between the corresponding projective spaces have closed images. Its pullback to the spectrum parameters is proper. Here is an exact witness:

$$\alpha = (3/2, 5/3, 7/4), \quad e = (1/3, -2/5, 3/7), \quad d = (2/3 + i/5, 3/5 - i/7, -4/7 + i/3), \quad g = \mathbf{1}.$$

After clearing scalar denominators the forms are

$$\begin{aligned} F = {} & -3365t^2u^2 - 14400t^2u + 2321t^2 + 10020tu^2 - 7612tu \\ & - 3060t + 1485u^2 - 1320u - 441, \\ G = {} & -20735t^2u^2 - 44520t^2u + 4613t^2 + 96460tu^2 - 49896tu \\ & + 10780t + 20335u^2 + 41160u - 4213. \end{aligned}$$

Their exact gcd is one and both have full bidegrees. Write the primitive-normalized resultant quotient as

$$\begin{aligned} & 65532894182001t^6 - 846134370496944t^5 + 2404104611159391t^4 \\ & + 2245261288159824t^3 - 463362875655073t^2 - 249992387249952t - 20785328196111. \end{aligned} \tag{35}$$

Denote the displayed polynomial by $S_{\text{prim}}(t)$. For precisely the integer F, G printed above, the ordinary Sylvester determinant satisfies

$$\operatorname{Res}_u(F, G) = 100(1 + t^2) S_{\text{prim}}(t). \tag{35a}$$

Thus the literal quotient in (18) is $100 S_{\text{prim}}$, not S_{prim}. Dividing out this positive integer content leaves the roots, multiplicities, degree and common-component witness unchanged. The universal construction continues to use the ordinary quotient (18); primitive normalization is only a convention in this example. The loop imaginary part is $1/1225$. No boundary divisor is a common component. The same property was separately checked with mixed weights $(2, -1, 3)$. Thus the needed open set is nonempty both for the primal slice $g = \mathbf{1}$ and for independent weighted data. Exact gcd/resultant and degree certificates are included in results/hexagon_spectrum_probe.json.

6.2 Independent dual parameters

The inverse map preserves the structured operator space. Parametrize a structured self-adjoint R by a real symmetric diagonal block G, three real scalar off-diagonal coefficients c^R, and three complex coefficients d^R. Its weighted data are

$$\alpha'_i = 2G_{ii}, \quad e'_{ij} = 2(G_{ij} + c^R_{ij}), \quad d'_{ij} = 2d^R_{ij}, \quad g = G\mathbf{1}. \tag{36}$$

This is a linear isomorphism between fifteen real coordinates. For example

$$G_{12} = \frac{g_1 + g_2 - g_3 - G_{11} - G_{22} + G_{33}}{2},$$

with the other two off-diagonal entries recovered cyclically. The remaining coefficients follow immediately. An exact Jacobian computation in the recorded coordinate order gives determinant 8192. Operator inversion is birational where defined. Consequently independent original couplings produce generic weighted dual data; the dual exceptions cannot be hidden identically by a parameter constraint.

6.3 No stationary zero on the pole locus, generically

Fix a support of size $s \geq 2$, nonzero supported components, generic g, and impose $\mathbf{1}^T r = 1$, $g^T r = 0$. If this set is nonempty, its radial dimension is $s-2$; its phase dimension is $s-1$. The total is $2s-3$. When g is constant and nonzero on the support the locus is empty, and a singleton cannot lie on it because $g_i \neq 0$.

At each such point, the value of $p = r^T B(q) r$ and its $2s-3$ tangential first derivatives give $2s-2$ independent linear conditions on the twelve numerator coefficients. To see independence explicitly:

- Vary $\mathrm{Im}(d_{ij} q_j \bar{q}_i)$ on a spanning tree, while adjusting e_{ij} to leave B fixed at the point. This changes the $s-1$ phase derivatives independently without changing the value or radial derivatives. The incidence-matrix rank is $s-1$, since all supported $r_i \neq 0$; for the tree $12, 23$, a determinant is $r_1 r_2^2 r_3$.
- Arbitrary real symmetric changes of B are available from α, e. In particular $\delta B = (\ell \mathbf{1}^T + \mathbf{1}\ell^T)/2$ gives $\delta p = \ell^T r$ on $\mathbf{1}^T r = 1$. Its value and the $s-2$ independent radial derivatives can be prescribed independently.

The incidence set of stationary zeros therefore has dimension at most $12 + (2s-3) - (2s-2) = 11$ for fixed g, or at most fourteen in the full fifteen-dimensional weighted parameter space. It is semialgebraic; its projection has no larger dimension and is contained in a proper algebraic set. There are only finitely many supports and coordinate charts. Thus algebraically generic weighted data have no such stationary zero. These elementary semialgebraic projection/dimension facts are the only dimension-theoretic input [BPR06].

A second verification of this particular genericity mechanism avoids the incidence count. Hold g fixed and replace p by $p + \eta(\mathbf{1}^T r)^2$, an allowed independent direction $\delta\alpha_i = 2\eta, \delta e_{ij} = 2\eta, \delta d_{ij} = 0$. On the normalized pole locus this adds the constant η without changing its stationary points. A semialgebraic differentiable function has finitely many critical values: its critical set has finitely many semialgebraic path-connected components and the function is constant on each by differentiation along paths. Hence an independent transcendental η cannot make a critical value zero. This also establishes that the pole-zero exception is not identically imposed by the model.

6.4 Simultaneity

The original coefficient space is irreducible affine fifteen-space. The primal conditions are proper by the explicit witness and the surjective map $(a, b, c, d) \mapsto (a, b+c, d)$. The dual conditions pull back properly by (36) and birational inversion. A finite union of the proper exceptional sets is still proper. No assumption that all these tests happen to hold in one arbitrary numerical example is needed. Algebraically independent formal coefficients avoid all of them simultaneously.

7. Universal singular completion by a positive infinitesimal

Order the coefficient coordinates as

$$(a_1, a_2, a_3, b_{12}, b_{13}, b_{23}, c_{12}, c_{13}, c_{23}, \mathrm{Re}\, d_{12}, \mathrm{Im}\, d_{12}, \mathrm{Re}\, d_{13}, \mathrm{Im}\, d_{13}, \mathrm{Re}\, d_{23}, \mathrm{Im}\, d_{23}).$$

Let $M_j(K)$ be the coefficient monomial, so $f = \sum_{j=1}^{15} \kappa_j M_j$. For an imaginary part of d, this monomial is $-\mathrm{Im}\, K_{ij}^2$. Work in the iterated real closed algebraic Puiseux field

$$F = \mathbb{R}\langle \epsilon_0 \rangle \langle \epsilon_1 \rangle \cdots \langle \epsilon_{15} \rangle, \qquad 0 < \epsilon_{15} \ll \cdots \ll \epsilon_1 \ll \epsilon_0 \ll 1. \tag{37}$$

A later infinitesimal is smaller than every positive element of the preceding field. These are sixteen independent formal variables, not sixteen numerical values chosen by a tolerance rule. Define

$$f_\epsilon = f + \epsilon_0 (\mathrm{tr}\, K)^2 + \sum_{j=1}^{15} \epsilon_j M_j(K). \tag{38}$$

Thus a_i and b_{ij} each acquire $2\epsilon_0$, followed by the coordinate's own ϵ_j. Other coordinates acquire their own ϵ_j only. The resulting fifteen coordinates are algebraically independent over $\mathbb{R}(\epsilon_0)$: they are independent variables plus fixed translations. Section 6 consequently applies without a genericity oracle.

On $\mathcal{D}$, $r_i^2/2 \le 1/2$, and every other coefficient monomial has absolute value at most $1/4$, because $|K_{ij}|^2 \le r_i r_j \le 1/4$. Therefore

$$\left|\sum_{j=1}^{15} \epsilon_j M_j(K)\right| \le \frac{9}{2}\epsilon_1 \ll \epsilon_0. \tag{39}$$

The bound is deliberately conservative. Taking minima and using Section 5 gives

$$|m_\epsilon - m - \epsilon_0| \le \frac{9}{2}\epsilon_1. \tag{40}$$

For $m < 0$ the negative real constant dominates the deformation. For $m > 0$ it exceeds $2\epsilon_0$. For $m = 0$, (40) places m_ϵ strictly between zero and $2\epsilon_0$. This proves all tests in (3), with no exchange of an uncontrolled limit and optimization.

Every minimizing retained matrix is positive semidefinite with trace one, so all its entries are bounded by real constants. Successive residue maps exist, are order-preserving ring homomorphisms, and must be applied in the order $\epsilon_{15}, \epsilon_{14}, \dots, \epsilon_0$ [LRT25]. Principal minors, trace and determinant survive these residues. Thus $K_0 \in \mathcal{D}$, and polynomial evaluation together with (40) gives $f(K_0) = m$.

Do not set the original singular operator inverse to its limit separately. The inverse may have poles. Construct the bounded physical matrix first, then take its residue. This also covers merging roots and stationary continua.

8. Finite exact operations and witness recovery

The stationary-candidate stage uses one homogeneous four-by-four resultant per spectrum (evaluated in the needed projective charts), univariate roots of degree at most six, common roots of specialized quadratics, the face square roots, finite arithmetic, signs and small matrix operations. Multiple real roots and all projective charts are retained. Ordered Thom encodings and univariate sign determination work over general real closed fields [BPR06, Algorithm 10.14 and its surrounding root/sign constructions]. Rational-interval bisection is not an adequate substitute in a non-Archimedean field.

The residue operation in (3) is also finite and explicit. Suppose a selected bounded algebraic number x lies over $E\langle\epsilon\rangle$, where E is the preceding real closed field.

1. Compute a nonzero annihilating polynomial $P(\epsilon, X) \in E[\epsilon, X]$ from the finite algebraic root tower, using field norms and clearing denominators. Equivalently eliminate in that already zero-dimensional finite algebraic tower, with the selected nonzero denominators localized. This is not elimination of the original field-orientation problem.
2. Divide the largest common power of ϵ, obtaining $Q(X) = P(0, X) \neq 0$.
3. Isolate and order the distinct real roots $q_1 < \cdots < q_s$ of Q in E. Since x is bounded, its residue is a root of Q.
4. Compare the selected x with the midpoints $(q_i + q_{i+1})/2$ in the ordered extension. These comparisons uniquely identify its residue. A midpoint is not a root of Q, so no ambiguity arises even when several branches coalesce at a repeated root.

There are finitely many finite-degree operations at each of the sixteen levels. Residue annihilators can have higher degree than six; the sextic bound applies to the phase-elimination stage, not to a minimal polynomial of the final energy or to every polynomial in a singular-input implementation. This selected-root procedure requires neither simple roots nor a claim that an arbitrary multivariate limit algorithm handles singular solutions. Carry the physical root selection through the computation; do not take limits of an unfiltered complex elimination set.

Finally construct doublets from the resulting K_0. Choose i with $K_{ii} > 0$ and set $a_j = K_{ij}/\sqrt{K_{ii}}$. The Schur complement $H_{jl} = K_{jl} - \bar{a}_j a_l$ is positive semidefinite of rank at most one. If $H = 0$, take the second row to be zero. Otherwise choose k with $H_{kk} > 0$ and set $b_j = H_{kj}/\sqrt{H_{kk}}$. Then

$$\Phi_j = (a_j, b_j)^T, \qquad \Phi^\dagger \Phi = K_0. \tag{41}$$

This recovers an exact minimizing field direction, and hence a negative or zero direction in the respective cases. It is not a classification of every zero orbit when the zero set is continuous.

8.1 Uniform finite-input construction and its F002 refinement

Proposition (finite-input endpoint). Fix any original fifteen-coordinate input κ and exact primitive interfaces for the ordered tower. There is a uniform finite code-producing construction of a nonempty list of physical candidate records. Its interpreted distinct scalar values are precisely $\mathcal{C}_\epsilon$, with cardinality at most thirty. Its selected record has value m_ϵ, its four sign tests satisfy (3), and its successive residue execution returns stage-zero codes for m, a minimizing physical Gram matrix, and normalized doublets attaining m. Every retained minimizing matrix of the semantic construction, with its original sign/gauge choices, has a code in a uniform finite covering list whose residue execution and pivot reconstruction attain the same original minimum. No genericity, nonsingularity, physicality or boundedness promise is imposed on the original input.

Here "code-producing" is relative to semantic exact primitives. It is not backend-independent serialization or an extracted executable implementation.

Construction and refinement proof. F002 makes the composition explicit. CoefficientCod en has only rational constants, the fifteen original input coordinates, and n formal infinitesimal generators as scalar leaves, together with addition, multiplication, negation and inverse. Coden adds finite selected-root nodes and square-root nodes. A root node contains a finite coefficient-code list and a natural-number index into the primitive root list; it does not insert an arbitrary ambient scalar as a constant. Code0 has no infinitesimal leaf because its generator index type is empty. Inverse is totalized in the syntax; the retained construction proves the denominators needed by the mathematical formulas nonzero rather than using division by zero as a mathematical step.

First, compile the canonical deformation (38), both parameter maps (22)–(23), and the singleton, pair and full-support candidate formulas into this syntax. Finite root enumeration produces all root-index codes and the relevant square-root signs. Exact equality/order tests perform retention and finite minimum selection. The stronger formal genericity exclusions hold for this canonical deformation for every κ; see Section 9.1. Interpretation commutes with the compiled arithmetic, operator reconstruction and physical filter. Consequently actu alCodeProgram_values identifies the interpreted value set with the semantic candidate set, actualCodeProgram_physical proves physicality and energy for each retained record, and actualSelectedRecord_value proves that literal finite-list selection has value m_ϵ. Together with (33) and (40), signPrograms_spec gives all four sign equivalences. The thirty bound is a bound on distinct values, not on the raw list of codes or gauge representatives.

Second, residue execution consumes the code that was actually generated. Finite multiplication presentations and their characteristic-polynomial annihilators produce coefficient programs; denominators are cleared and the maximal common infinitesimal power is stripped. Root-index codes for the nonzero specialized polynomial are generated in the preceding stage. In the formal implementation, finite exact comparisons select the root closest to the source value in the next ordered field. If q is the residue, the distance to q is infinitesimal relative to the preceding field, whereas the distance to each different preceding-field root is bounded below by a positive preceding-field element. Hence the selection uniquely recovers the residue. This is equivalent in outcome to the midpoint description above; neither method assumes simple roots. Typed distance requests retain both operand codes. codedResidueStageProgr am_correct proves one-stage success, and structural recursion gives sixteen-stage success. Scalar and physical matrix bounds are proved from the input and retention, not supplied as additional hypotheses to the central endpoint. actualValueCode_success identifies the final scalar with m; actualMatrixCode_spec gives a physical Gram matrix with energy m.

Third, the literal pivoted square-root construction (41), also with finite root-index codes, gives actualFields_spec: total norm one, the computed Gram matrix, and quartic value m. The normalized fields therefore witness the negative or flat alternatives when their corresponding sign test holds. Finally, every_old_retained_minimizer_coded_execution strengthens this selected record statement to every semantic retained minimizer. Its covering list can contain extra nonphysical sign combinations; physicality is asserted for entries representing the retained minimizing matrices, not for every covering list entry. This statement does not classify all real minimizing or zero orbits in a continuous exceptional locus.

These components are assembled in W3.FiniteInput.uniform_finite_input_exact_criterion; the field-domain sign statement for arbitrary real and complex couplings is W3.FiniteInput.arbitrary_complex_finite_input_criterion. The theorem parameter ResidueOracles supplies coefficient equality at each rational-tower stage and finite real-root enumeration at each ordered stage. Arithmetic and order are interpreted in the semantic ordered fields; some equality/order/list-selection choices use classical choice at runtime. Thus finite input provenance and refinement close the F001 composition gap, but do not establish a concrete representation-level exact solver or remove classical choice from the program.

9. Scope, provenance and verification limits

The theorem concerns only the tree-level quartic form (1). If $m = 0$, the quadratic terms can still be negative along a flat quartic direction. No claim is made here that arbitrary $V_2 + V_4$ is bounded below, that a phenomenologically acceptable neutral vacuum is global, or that radiative corrections preserve the classification. Conversely $m > 0$ is strict quartic positivity.

9.1 Analytic and formal auxiliary domains

Let G denote the broader regular domain of the analytic four-projective-chart and specialized-gcd procedure in Sections 2 and 6. Let G′ denote the stronger open domain of the formal affine candidate API: it additionally excludes projective-boundary roots and nonunique linear recovery fibers in its affine chart. These domains must not be identified. F002 proves projective-root equivalence on G′ and proves that both canonical deformed inputs lie in G′ for every original coefficient vector. Therefore the central all-input theorem is unrestricted, although the broader undeformed all-G auxiliary interface and its general intersection-multiplicity count are not separately packaged in full in Lean. The optional coefficient-evaluation formula (28) is also not a separate literal formal endpoint; the formal physical filter uses the proved purity identity. Sections 2–6 retain the broader analytic proof; Section 8.1 states the narrower auxiliary interface actually used by the formal endpoint.

9.2 Verification and practical limitations

The repository's F002 release-status record, dated 9 October 2026, reports fresh compilation of 190 local Lean modules and strict fresh replay of 4,545 declarations (3,088 theorems) by LeanChecker using the same Lean kernel. Its source freeze is SHA256 7fc34595fdb0a21778 cf000dfac1484a4d6dd9c0e9dc3aff7ad0fce5f1af73c5. It records three repaired independent AI audit rounds with four reports per round and adjudication, with no confirmed or open potentially load-bearing errors recorded. These are source-bound historical validation results, not a fresh build or new adversarial audit of this revised prose.

The local scientific declaration gate excludes custom scientific axioms, sorry/admit, unsafe/partial W3 declarations, native-decide shortcuts and compiler-trust bypasses. The allowed foundational axioms are propext, Classical.choice and Quot.sound. LeanChecker uses the same kernel, not an independent proof-kernel implementation. Runtime declaration skipping is disclosed separately from the stricter local W3 gate; pinned external dependency artifacts are fingerprinted/reused rather than all rebuilt from source. A successful remote GitHub Actions run is a separate fact and is not inferred from the local release-status record.

The algebraic genericity, root/sign and bounded-residue arguments are proof ingredients, not conclusions drawn from floating-point agreement. The prototype uses tolerances and local polishing and is only diagnostic. Symbolic scripts check named identities and finite examples.

The historical diagnostic replay wrapper could accept stale outputs and must not be an assurance gate; the separate hardened wrapper requires fresh outputs and strict comparisons. Its recorded reproduction of eight finite suites does not prove universal stationarity, genericity or residue coverage.

No verified rational/algebraic backend, runnable extracted exact solver, backend-independent portable root-code serialization, practical runtime advantage, human peer review, external endorsement or absolute priority certification is claimed. The source-based priority assessment was bounded to the material inspected on 9 October 2026; this manuscript revision does not refresh that literature survey. Independent AI review and kernel checking must remain distinct from human scholarly review.

References used at their stated scope

[BPR06] S. Basu, R. Pollack and M.-F. Roy, Algorithms in Real Algebraic Geometry, second edition, 2006. Real closed fields, semialgebraic dimension and univariate root/sign representations, especially Algorithm 10.14. The singular residue procedure in Section 8 is proved here rather than inferred from a simple-zero-only multivariate limit algorithm.

[LRT25] A. Lerario, L. Rizzi and D. Tiberio, Quantitative approximate definable choices, Mathematische Annalen 392 (2025), 1289ff. Algebraic Puiseux towers, bounded residues and the order of successive limits; Definitions 4, 23, 25 and Remarks 26–27.

[MN15] M. Maniatis and O. Nachtmann, Stability and symmetry breaking in the general three-Higgs-doublet model, arXiv:1408.6833; JHEP 02 (2015) 058, with erratum JHEP 10 (2015) 149. The inspected arXiv text includes the rank-one stationary treatment. Gram geometry and the general stationary-equation method are prior work, not claimed as new here.

The source and bounded equivalent-prior assessment, including recent Weinberg model work, is preserved in research/SOURCES.md of the anchored repository. Formal status and scope are recorded in verification/RELEASE_STATUS.json and docs/SCOPE.md; confirmed corrective history is in docs/ERRATA.md.